



**REPORT ON AN IN-SITU INTEGRITY TEST
(NON-DESTRUCTIVE) OF COMPRESSIVE
STRENGTH OF STRUCTURAL ELEMENTS**

BY

**LAGOS STATE MATERIALS TESTING LABORATORY
OJODU BERGER, LAGOS.**

ON

AN EXISTING SITE

(Botanical Garden Road, Ebute Metta)

FOR

THE CHIEF MAGISTRATE COURT LAGOS

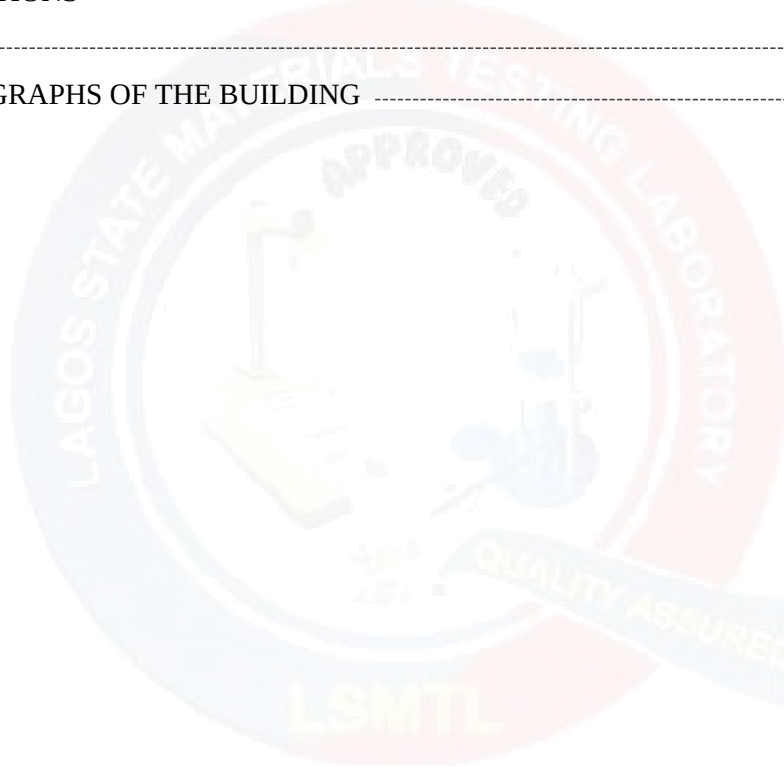
BOTANICAL GARDEN ROAD, EBUTE METTA, LAGOS, LAGOS MAINLAND, LAGOS
(0906 833 5593)



04TH SEPTEMBER, 2026.

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EXECUTIVE SUMMARY

In situ Integrity Test (Non-Destructive) of compressive strength of structural members of an existing 2-floor building belonging to THE CHIEF MAGISTRATE COURT LAGOS, at Botanical Garden Road, Ebute Metta, Lagos.

The Non-Destructive Integrity Test was carried out on the 04TH SEPTEMBER, 2026 with the intention to determine the residual compressive strength of concrete components of the structural members considered to be critical to stability, robustness and general safety of the entire structure in its present state.

The visual inspection revealed structural condition indications across key load-bearing members. Non-Destructive test analysis shows varying compressive strength bands across tested structural members. Where as-built structural drawings were unverified, an assumed design compressive strength of 25 N/mm² was applied as the statutory baseline benchmark.

In view of the above, it is advised that a qualified structural engineer and relevant built-environment professionals should be engaged to proffer solutions to defects observed, provide technical guidance on deficient structural members identified, and further analyze the structural arrangement to guarantee the long-term stability, integrity, and serviceability of the structure.

1.0 INTRODUCTION

Non-Destructive Test (NDT), as the name implies, means that the material under test is not damaged during testing. Direct measurement of the strength of concrete involves destructive stresses and cannot be used for determining the quality of already cast concrete. It is for this reason that direct destructive methods are not employed in determining the strength of in-situ concrete. As a result, indirect ultrasonic pulse velocity methods are used.

In compliance with the statutory Non-Destructive testing mandate, a Non-Destructive compressive strength test (Structural Integrity Test) was conducted on the structure located at Botanical Garden Road, Ebute Metta, Lagos. Testing was executed with the Portable Ultrasonic Non-Destructive Digital Indicating Tester (PUNDIT) in strict accordance with BS 1881-203:1999, BS EN 12504-4:2021 and ASTM C597.

Project Name	Botanical Garden Road, Ebute Metta
Site Address	Botanical Garden Road, Ebute Metta, Lagos
Local Government Area	Lagos Mainland
State	Lagos
Structural System	-
Assumed Design Strength	25 N/mm ² (BS 8110: Part 1 1997 / BS EN 1992)
Ultrasonic Tests Logged	0

2.0 PURPOSE OF INVESTIGATION

The purpose of the investigation is to:

1. Determine the present status of the buildings from the outcome of the visual and Non-Destructive test of the tested buildings to ascertain their state with respect to BS EN 12504-4:2021 and ASTM C597-09.
2. Determine the ongoing concrete strength of the building structure with respect to BS CODE 8110: Part 1 1997.
3. Provide engineering advice based on the visual and Non-Destructive test conducted.
4. To comply with government statutory requirements and ensure public safety.

3.0 LITERATURE REVIEW

Ultrasonic Pulse Velocity (UPV) testing is a non-destructive testing technique used for assessing the quality, uniformity, and internal condition of hardened concrete. The method operates by transmitting high-frequency ultrasonic waves through concrete and measuring the travel time between transmitting and receiving transducers.

The pulse velocity is calculated using the relationship: $UPV = L / t$

Where: L = Distance between transducers (mm), t = Pulse transit time (us)

The testing procedure and interpretation of results are conducted in accordance with international standards including:

- ASTM C597 - Standard Test Method for Pulse Velocity Through Concrete (2020)
- BS EN 12504-4:2021 - Determination of ultrasonic pulse velocity in concrete
- BS 1881-203:1999 - Recommendations for measurement of velocity of ultrasonic pulses in concrete
- ACI 228.2R - Report on Nondestructive Test Methods for Evaluation of Concrete in Structures

Estimated compressive strength (E.C.S) values in this report are derived from the laboratory's fixed ultrasonic pulse-velocity calibration curve $f_{cu} = 8.96 \times V - 7.97$ (f_{cu} in N/mm², V in km/s), established by least-squares regression over the laboratory's reference control pairs (2.9, 19), (3.9, 25), (4.0, 27), (4.2, 29) and (4.4, 34) and valid over the range 2.0 - 5.0 km/s. BS 1881-203:1999 notes that no unique velocity-strength relationship exists for all concretes; the curve above is the documented calibration applied by this laboratory for Nigerian site concrete and is applied uniformly to every result in Section 5.0. Velocities outside the calibrated range are reported without an E.C.S estimate rather than extrapolated.

4.0 FIELD WORK

The field work was executed on 27th April, 2026. Visual testing was carried out on the building to ascertain any structural defects (cracks, differential settlement, spalling, honeycombing, hogging, and sagging). Standard Portable Ultrasonic Non-Destructive Digital Indicating Testers (Pundit PL-200) were employed for estimating compressive strength of hardened concrete, and electromagnetic rebar locators (Profoscope) were used to ascertain rebar positioning, concrete cover, and bar diameters.

During the investigation, several key operational factors were accounted for:

- Surface conditions, ambient temperature, and moisture content of the concrete.
- Path length, transducer transmission configuration (direct, semi-direct, indirect).
- Concrete stress state and influence of embedded reinforcing steel bars.

EQUIPMENT STATUS CHECK

Name of Equipment	Test Equipment ID	Transducer	BNC Cables	Display on Machine	Battery Status
UPV (Pundit PL-200)	LSMTL-UPV-001-0001	54 kHz (OK)	1.5m BNC (OK)	Digital LCD (OK)	75% (OK)

The test was conducted in the presence of designated Lagos State Materials Testing Laboratory staff:

- Engr Abiola-Balogun A.S. (R.41589)
- Engr Akinfeleye Kayode (R.60435)
- Engr. Tech. Aderibigbe Ridwan (R.78839)
- Tosin Salami (NDT Field Technician)

This test forms part of the Lagos State Government statutory mandate to mitigate building failure and prevent structural collapse within the state.

4.1 VISUAL TEST

- Structures are in active occupation at time of inspection.
- Superficial hairline cracking observed on selected ground floor columns.
- Localized dampness and floor finish wear observed at ground floor utility zones.
- No visible global deflection, buckling or foundation subsidence observed.

4.2 METHODOLOGY

NON-DESTRUCTIVE CONCRETE STRENGTH AND REBAR DETERMINATION:

A pulse of longitudinal vibrations is produced by an electro-acoustical transducer held in contact with the concrete surface using a certified coupling gel. The pulse propagates through the concrete, is received by a second transducer, and electronic timing circuits measure the transit time (t). The pulse velocity V is computed as $V = L / t$. Ultrasonic indications are correlated to in-situ compressive strength f_{cu} via the statutory LSMTL calibration model: E.C.S: $f_{cu} = 8.96 \times V - 7.97$ (f_{cu} in N/mm², V in km/s; valid 2.0 - 5.0 km/s)

4.3 REINFORCING BAR (REBAR) ASSESSMENT

During testing, electromagnetic rebar locators (Profoscope) were employed to scan the concrete cover depth, verify steel rebar orientation, and record estimated bar sizes embedded in members.

EQUIPMENT / REBAR ASSESSMENT TABLE (Profoscope LSMTL/Prof/026/003)

S/N	STRUCTURAL MEMBER	MAIN BAR (MM)	LINKS (MM)	SPACING (MM)	COVER DEPTH (MM)
1	COLUMN	16	10	300	50
2	BEAM	16	10	300	30
3	SLAB	12	10	-	30

NOTE: This assessment is non-destructive and does not replace construction reinforcement design drawings.

5.0 ANALYSIS OF TEST RESULTS

The structural elements were tested across multiple grid stations (Points A, B, C) using calibrated transducers. Compressive strength is graded against the statutory baseline of 25.0 N/mm². Elements meeting or exceeding 25.0 N/mm² are classified as GOOD; elements below 25.0 N/mm² are classified as POOR.

SUMMARY OF TEST ANALYSIS

STRUCTURAL MEMBER	NUMBER TESTED	LOCATION	NO OF POINTS TAKEN
COLUMN	44	Ground Floor	132
COLUMN	32	First Suspended Floor	96
BEAM	3	First Suspended Floor	9
SLAB	3	First Suspended Floor	9

DETAILED STRUCTURAL ELEMENT TEST READINGS

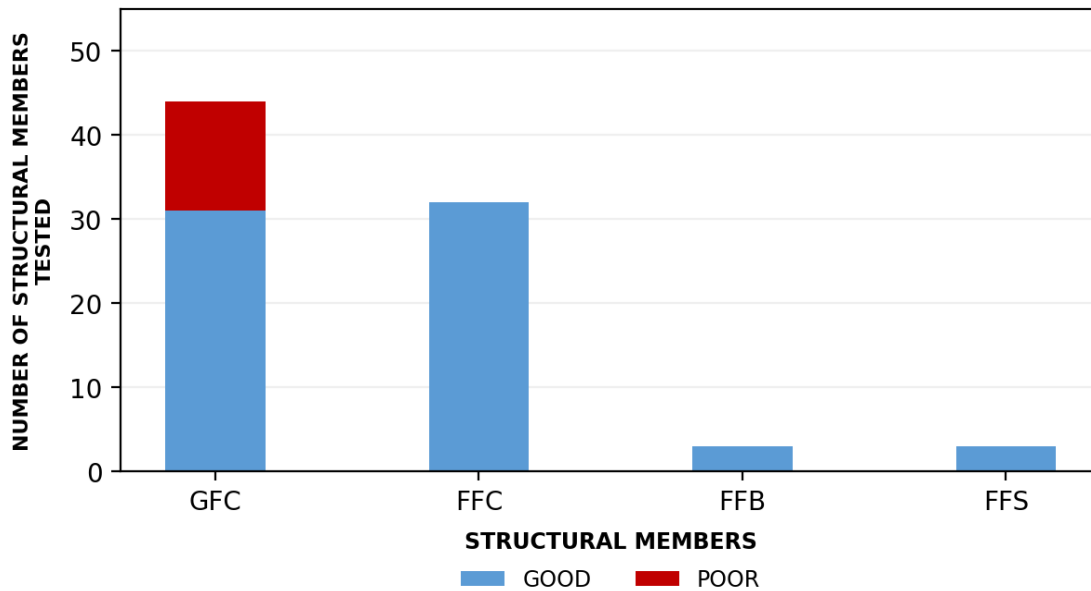
Structural Element	Pt	PATH (MM)	TRANSIT (US)	VELOCITY	E.C.S	AVG STRENGTH	REMARK
COLUMN 1	A	120	31.2	4.0	27.0	27.0	GOOD
-	B	120	32.2	3.9	25.0	-	-
-	C	120	30.1	4.2	29.0	-	-
COLUMN 2	A	120	30.9	4.1	28.0	17.8	POOR
-	B	120	48.9	2.6	12.0	-	-
-	C	120	44.7	2.8	13.3	-	-
COLUMN 3	A	120	32.4	3.9	25.0	27.7	GOOD
-	B	120	29.2	4.3	33.0	-	-
-	C	120	32.4	3.9	25.0	-	-
BEAM 1	A	120	32.6	3.9	25.0	27.0	GOOD
-	B	120	30.3	4.2	29.0	-	-
-	C	120	31.4	4.0	27.0	-	-
SLAB 1	A	120	30.6	4.1	28.0	26.7	GOOD
-	B	120	32.4	3.9	25.0	-	-
-	C	120	31.7	4.0	27.0	-	-

SUMMARY OF TEST RESULTS

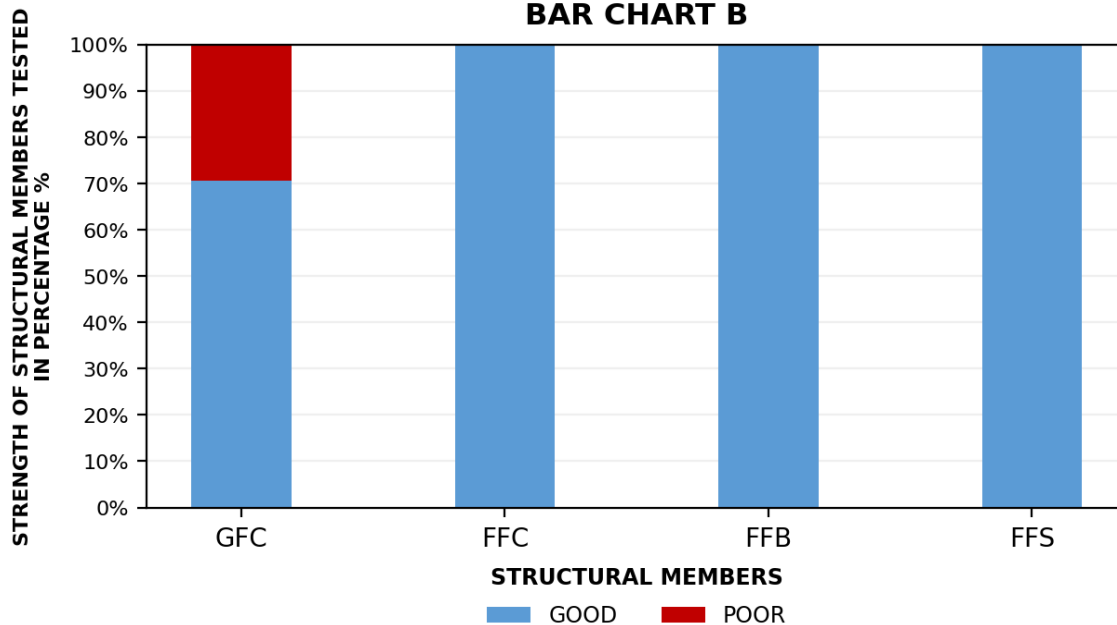
STRUCTURAL MEMBER	LOCATION	GOOD (NO)	GOOD (%)	POOR (NO)	POOR (%)
Column	Ground Floor	31	70.5%	13	29.5%
Column	First Suspended Floor	32	100.0%	0	0.0%
Beam	First Suspended Floor	3	100.0%	0	0.0%
Slab	First Suspended Floor	3	100.0%	0	0.0%

**BAR CHART SHOWING SUMMARY OF TEST RESULTS
(BOTANICAL GARDEN ROAD, EBUTE METTA)**

BAR CHART A



BAR CHART B



**CHART ILLUSTRATING THE NUMBER OF STRUCTURAL MEMBERS TESTED*

**CHART ILLUSTRATING THE PERCENTAGE OF STRENGTH OF STRUCTURAL MEMBERS TESTED*

Legend: GFC: Ground Floor Column | FFC: First Floor Column | FFB: First Floor Beam | FFS: First Floor Slab

6.0 RECOMMENDATIONS

Based on the purpose of investigation, the outcome of the Visual and Non-Destructive Test carried out on the building which reveals the present status of the structural members as described in the visual test and depicted in the analytical bar charts, the following statutory recommendations are formally issued:

- Engage a COREN-registered structural engineer to carry out detailed structural calculations and remediation designs for all structural elements categorized as POOR.
- Perform localized pressure grouting with low-viscosity structural epoxy or high-performance micro-cement across all detected voids and cracks exceeding 0.3 mm.
- Perform non-destructive re-testing (UPV verification) following remedial grouting to certify that acoustic velocity has been restored to ≥ 3.75 km/s.
- Carry out geotechnical substructure probing to verify foundation bearing strata stability where settlement or dampness indications were observed.

7.0 CONCLUSION

The visual and structural integrity test was conducted in accordance with BS 1881: Part 201: 1986, BS EN 12504-4:2004, and BS EN 12504-4:2021.

1. The Non-Destructive Test analysis as shown in the summary tables and analytical charts illustrates the strength distribution across the tested structural elements.
2. However, it is imperative to state clearly that non-adherence to the recommendations excludes the laboratory of any responsibility regarding ongoing structural performance.

NOTE: The test benchmark assumed 25 N/mm² as the characteristic compressive strength of the structural members; however, a substructure probe is required to ascertain the integrity of the building foundation.



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Date Certified: 04TH SEPTEMBER, 2026 | Lagos State Materials Testing Laboratory*

APPENDIX PHOTOGRAPHS OF THE BUILDING

Photographs depicting the tested structure, visual observations, and in-situ ultrasonic testing in progress:

Photographs from the field investigation are archived in the digital evidence vault.